

IP Addressing in Detail

Lecture 4 | Hinglish Notes with Diagrams

Ye document **IP Addressing** par lecture notes ko clean, point-wise format me Hinglish mein explain karta hai, saath me har concept ke diagrams bhi hain: IPv4 & IPv6 structure, Network ID / Host ID, Address Classes (A-E), subnet mask basics, loopback address, broadcast address aur real-world (ISP) IP examples.

1. IP Address Kya Hota Hai?

IP Address (Internet Protocol Address) ek unique logical address hota hai jo kisi bhi network se jude hue har device ko diya jata hai, taaki us device ko identify kiya ja sake aur wo doosre devices ke saath communicate kar sake. Ye bilkul postal address jaisa kaam karta hai — internet par data packets ko sahi jagah tak pahunchane ke liye.

Aaj IP addressing ke do standard versions use hote hain:

- **IPv4** – 32-bit address (purana hai, lekin abhi bhi sabse zyada use hota hai)
- **IPv6** – 128-bit address (naya hai, IPv4 ke addresses kam pad rahe the isliye banaya gaya)

2. IPv4 Addressing (32-bit)

IPv4 ek **32-bit** ka binary number hota hai, jo generally **dotted decimal notation** me likha jata hai — 32 bits ko 4 equal parts me todte hain, har part 8 bits ka hota hai, jise **octet** kehte hain. Har octet ko decimal number (0–255) me convert karke dots se separate kiya jata hai.

32-bit IPv4 Address = 4 x 8-bit Octets



Example: 192 . 168 . 1 . 5

Fig 1: 32-bit IPv4 address ka structure

IPv4 ke kuch important range facts:

- Minimum address: **0.0.0.0**
- Maximum address: **255.255.255.255**
- Total possible addresses = $2^{32} \approx 4.29$ billion
- **Loopback address:** 127.0.0.0 se 127.255.255.255 tak — ye reserved hota hai taaki device khud ko refer kar sake (jaise 127.0.0.1). Isse apne hi local machine ke network stack ko test karne ke liye use karte hain.

3. IPv6 Addressing (128-bit)

IPv6 isliye introduce kiya gaya kyunki IPv4 ke ~4.3 billion addresses khatam ho rahe the. Ye ek **128-bit** ka address hota hai, jo decimal ki jagah **hexadecimal** form me likha jata hai.

- 128 bits ko **8 groups me baanta jata hai, har group 16 bits ka** hota hai ($8 \times 16 = 128$ bits).
- Har 16-bit group ko 4-digit hexadecimal number ki tarah likha jata hai, aur groups ko colon (:) se separate kiya jata hai.
- Hexadecimal digits use hoti hain: 0–9 aur A–F.

IPv6: 8 groups x 16 bits = 128 bits



Fig 2: 128-bit IPv6 address = 16 bits ke 8 hextets

Full-form IPv6 address ka example range:

Minimum: **0000:0000:0000:0000:0000:0000:0000:0000**

Maximum: **FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF**

Total possible IPv6 addresses = 2^{128} , jo IPv4 se bahut zyada bada address space hai.

4. Network ID aur Host ID

Har IPv4 address logically do parts me divide hota hai:

- **Network ID (NID)** — ye batata hai ki device kis network ka part hai.
- **Host ID (HID)** — ye us network ke andar ek particular device (host) ko identify karta hai.

Kitne bits Network ID ko milenge aur kitne Host ID ko, ye depend karta hai address ki **Class** par (Section 5 me explain kiya gaya hai) ya use ho rahe **subnet mask** par.

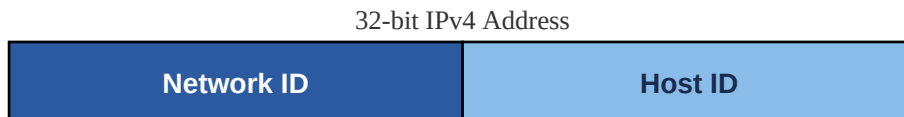


Fig 3: Har IP address Network ID + Host ID me split hota hai

Broadcast Address: Kisi network ke andar, jis address ke saare Host ID bits 1 set hote hain, wo **broadcast address** ke liye reserved hota hai — isse data ek saath us network ke saare hosts ko bheja jata hai (ye kisi single device ko assign nahi ki ja sakti).

5. Classful IP Addressing (Class A – E)

IPv4 ki original addressing scheme address space ko 5 classes me divide karti hai — A, B, C, D aur E — jo **first octet** ki value (aur isliye address ke shuru ke bits) par based hoti hain. Class A, B aur C normal host addressing ke liye use hoti hain (networks kam hote jaate hain lekin har network me hosts badhte jaate hain); Class D multicasting ke liye hoti hai; Class E future / experimental use ke liye reserved hai.

Class	1st Octet Range	Leading Bits	Default Mask	Network Bits	Host Bits	Purpose
A	1 – 126	0	255.0.0.0	8	24	Very large networks
B	128 – 191	10	255.255.0.0	16	16	Medium-sized networks
C	192 – 223	110	255.255.255.0	24	8	Small networks
D	224 – 239	1110	—	—	—	Multicasting
E	240 – 255	1111	—	—	—	Research / Future use

Fig 4: IP address classes ki summary table

Note: **127.x.x.x** (jo numerically Class A ke range me aata hai) normal hosts ke liye use **nahi** hota — ye poora Loopback address ke liye reserved hai.

5.1 Class ranges binary se kaise derive hote hain

Har class ko first octet ke fixed leading bit pattern se pehchana jata hai. Isi wajah se upar wali table me decimal ranges is tarah aate hain:

Class	Bit Pattern	Lowest Value	Highest Value	Rule
Class A	0 xxxxxxx	0 0000000 = 0	0 1111110 = 126	First bit fixed = 0
Class B	10 xxxxxx	10 000000 = 128	10 111111 = 191	First 2 bits fixed = 10

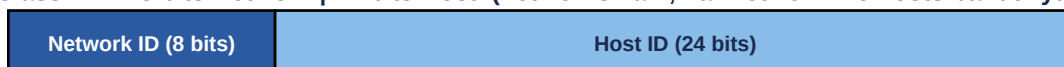
Class C	110 xxxxx	110 00000 = 192	110 11111 = 223	First 3 bits fixed = 110
Class D	1110 xxxx	1110 0000 = 224	1110 1111 = 239	First 4 bits fixed = 1110
Class E	1111 xxxx	1111 0000 = 240	1111 1111 = 255	First 4 bits fixed = 1111

(Note: 127 ko Class A ke usable range se skip kiya jata hai kyunki wo loopback ke liye reserved hai, isliye Class A ka usable first-octet range 1–126 bataya jata hai.)

5.2 Class-wise Network ID / Host ID split (diagram)

Kyunki total address hamesha 32 bits ka hota hai, Network ID ko zyada bits dene ka matlab (jaise Class A me) kam networks lekin har network me bahut zyada hosts — aur Class C me iska ulta hota hai.

Class A -> 8 bits Network | 24 bits Host (networks kam, har network me hosts bahut zyada)



Class B -> 16 bits Network | 16 bits Host (balanced split)



Class C -> 24 bits Network | 8 bits Host (networks zyada, har network me hosts kam)



Fig 5: Class A, B aur C ke liye Network ID / Host ID ka proportion

5.3 Har class me kitne networks aur kitne hosts

Class	Networks Available	Hosts per Network
A	$2^7 = 128$ (126 usable)	$2^{24} - 2 = 16,777,214$
B	$2^{14} = 16,384$	$2^{16} - 2 = 65,534$
C	$2^{21} = 2,097,152$	$2^8 - 2 = 254$

Host count me se 2 isliye minus karte hain kyunki jab saare host bits 0 hote hain to wo khud **Network Address** ban jata hai, aur jab saare host bits 1 hote hain to wo **Broadcast Address** ban jata hai — in dono me se koi bhi kisi actual device ko assign nahi kiya ja sakta.

6. Real-World Example: IP Address Kaise Padhein

Ek example lete hain — address **73.0.0.0**, jo ISP (Internet Service Provider, jaise notes me diye gaye Airtel / Idea / Vodafone type connections) use karte hain:

Address	73 . 0 . 0 . 0
First octet	73 -> 1-126 range me aata hai
Class	Class A
Network ID	73 (8 bits)
Host ID range	0.0.0 - 255.255.255 (24 bits)
Broadcast address	73.255.255.255

Isi tarah, **182.255.255.255** jaise address ka first octet 182 hai, jo 128-191 range me aata hai, isliye ye **Class B** ka address hai.

7. Quick Recap

- ✓ IPv4 = 32-bit address (4 octets), decimal me likha jata hai, dot se separate.
- ✓ IPv6 = 128-bit address (8 hextets), hexadecimal me likha jata hai, colon se separate.

- ✓ Har address = Network ID + Host ID.
- ✓ Class A/B/C = normal host addressing; Class D = multicast; Class E = reserved/experimental.
- ✓ Class A: 1-126, Class B: 128-191, Class C: 192-223, Class D: 224-239, Class E: 240-255.
- ✓ 127.x.x.x Loopback testing ke liye reserved hai, ye usable Class A network nahi hai.
- ✓ Broadcast address = network address jisme saare Host ID bits 1 set hote hain.